

Beyond the Mean: Nonparametric and Quantile Extensions of Feyrer, Mansur and Sacerdote (2017)

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Abstract

Feyrer, Mansur and Sacerdote (2017), hereafter FMS, use instrumental-variables (IV) regression to estimate the *average* causal effect of new oil and gas production on local wages in the United States. We extend their analysis along three dimensions the original paper cannot address. First, is the wage–fracking relationship linear? Local polynomial regression reveals a hump-shaped treatment response, not the straight-line FMS assume. Second, is the causal effect homogeneous across the wage-growth distribution? IV quantile regression shows that high-growth counties receive a larger immediate boom but a sharper subsequent correction: a boom–bust signature the mean conceals. Third, what does fracking do to spatial wage inequality? Unconditional RIF regression shows the 90th-percentile effect (\$31,000 per million dollars of production) is over six times the median effect (\$5,000): fracking widens the cross-county wage distribution in a way the original paper structurally cannot detect.

1. Introduction

Feyrer, Mansur and Sacerdote (2017) document how the US fracking boom propagated through local labour markets between 2005 and 2012. Using geological shale endowments as an instrument for county-level production, they find that each million dollars of new extraction raises wages by approximately \$80,000 within the county and \$257,000 within 100 miles, roughly trebling the naive OLS estimate. Every one of their estimates is a conditional mean; the paper is entirely silent on distribution.

Three gaps follow directly. The treatment response may be nonlinear: at extreme drilling density, congestion and factor-market crowding could reduce the marginal wage return. The causal effect may vary across the wage-growth distribution: the same geology instrument can be re-deployed at different quantiles to test this. And because fracking is highly spatially concentrated — the top ten per cent of counties account for 80 per cent of production — the mean cannot reveal whether gains are broadly shared or captured disproportionately at the top of the distribution. We address each gap in turn, using the same data and outcome variable as FMS throughout.

The remainder of the paper is organised as follows. Section 2 summarises the FMS paper, including its spatial mechanics. Section 3 describes our methods. Section 4 presents nonparametric results. Section 5 presents quantile regression results. Section 6 discusses implications and caveats. Section 7 concludes.

2. The FMS Paper

2.1 Research question and data

The paper asks how much of the value of new oil and gas production stays within the county and region where extraction occurs, and how those gains evolve over time and across industries. Production data come from Drilling Info (well-level output for every month and county, 2004–2014), valued at EIA spot prices to give a dollar measure of new production per county per year. Wages and employment come from the BLS Quarterly Census of Employment and Wages, disaggregated into eight industry groups. Income, including royalties and non-wage business receipts, comes from IRS Statistics of Income (available to 2012). All variables are scaled by one-year-lagged employment; all monetary values are in 2014 US dollars.

2.2 Empirical framework

The estimating equation is:

$$\Delta Y_{it} = \beta_1 \text{NewValue}_{it} + \beta_2 \text{NewValue}_{it-1} + \alpha_i + \omega_t + \varepsilon_{it}$$

where ΔY_{it} is the one-year change in wages (or employment) per lagged worker in region i in year t ; NewValue_{it} is the dollar value of production from wells that began producing in year t , scaled by lagged employment; α_i and ω_t are county and year fixed effects. Standard errors are two-way clustered by county and year.

Because firms choose where to drill, potentially targeting counties with favourable economic conditions, OLS estimates are downward-biased. FMS address this by instrumenting production using a play-year interaction: predicted county-level production derived from shale geology (play fixed effects) interacted with common annual shocks to drilling profitability. Since geology is fixed by nature and national energy prices are determined independently of any individual county, neither component of the instrument responds to local wage trends. First-stage F-statistics exceed 10 for all preferred specifications.

2.3 Spatial mechanics

A county-level analysis alone understates the full effect of fracking because workers and landowners often live in counties adjacent to drilling sites. FMS therefore aggregate production, income, and employment within concentric circles of increasing radius around each county centroid, from 20 to 200 miles, and re-estimate equation (1) at each distance. This is equivalent to constructing artificial commuting areas of varying sizes. The instrument is constructed analogously at each geographic level, so the causal interpretation is maintained throughout.

The mechanism operates through three channels. First, workers in mining and transport receive direct wage payments regardless of which county they sleep in. Second, landowners and mineral-rights holders, who may be dispersed well beyond the drilling site, receive royalties that flow through the IRS "other income" category. Third, increased local demand from higher incomes generates secondary employment and wage gains in construction, hospitality, and local government. Each of these channels radiates outward at a different speed and distance.

Figure 1 illustrates both the spatial structure of propagation and the magnitudes from FMS Table 1.

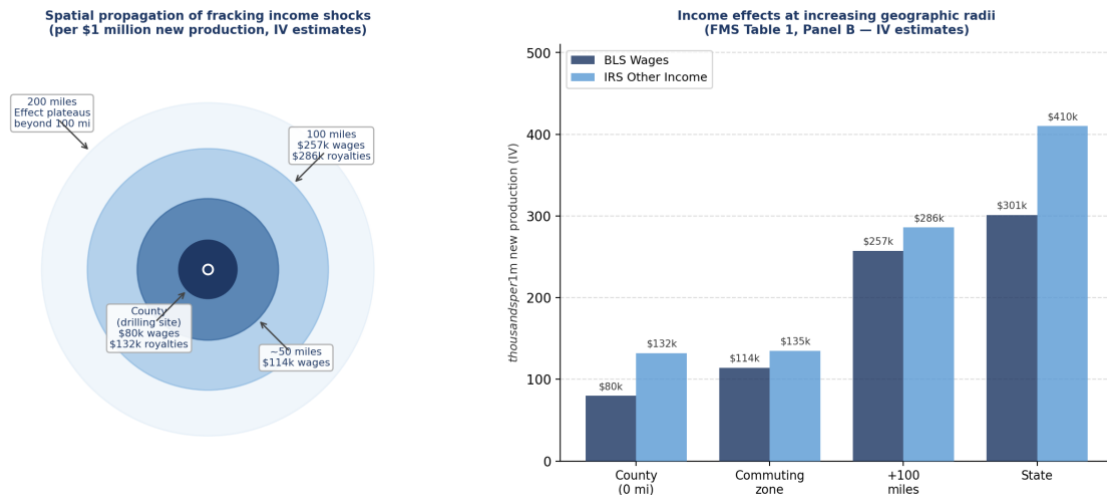


Figure 1. Spatial propagation of fracking income shocks. Left: concentric-circle structure of the geographic analysis, with IV-estimated wage and royalty effects per \$1 million of new production. Right: total wage and other income effects at each level of geographic aggregation (FMS Table 1, Panel B).

The key empirical finding from the spatial analysis is that effects are approximately flat between 0 and 50 miles, rise steeply between 50 and 100 miles as adjacent counties are drawn in, and then plateau beyond 100 miles. This 100-mile threshold corresponds broadly to commuting distance, beyond which workers are unlikely to be sourced from the local labour pool. Royalty and business income continues to drift upward past 100 miles, suggesting that mineral-rights ownership is more geographically dispersed than employment.

2.4 Main findings

At the county level, each million dollars of new production raises BLS wages by \$80,000 (IV) and generates 0.85 jobs; a further \$132,000 accrues as royalties and business income. Combined, roughly 20 per cent of production value is retained within the county. The regional multiplier is approximately three times larger: \$257,000 in wages and \$286,000 in other income within 100 miles, totalling 54 per cent of production value.

Wage effects are persistent: two-thirds of the initial increase remains two years after the production shock, driven by continued growth in non-mining sectors (transportation, services,

government). Mining wages dissipate within two years. Manufacturing shows a mild contemporaneous decline consistent with Dutch Disease. In aggregate, FMS attribute 640,000 additional jobs and a 0.43 percentage-point reduction in the unemployment rate during the Great Recession to new extraction.

3. Our Extensions: Methods

All methods absorb county and year fixed effects via iterative Gauss–Seidel within-group demeaning (Canay, 2011) before estimation, since standard quantile and nonparametric estimators do not support fixed-effects absorption directly.

Kernel density and CDF comparison contrasts wage-growth distributions for high-fracking counties (top decile of combined current and lagged production) against counties with zero production in both periods. A Kolmogorov–Smirnov test assesses statistical distinguishability.

Local polynomial regression fits a degree-1 Epanechnikov kernel smoother to the FE-demeaned wage residual on current production among fracking counties (trimmed at the 99th percentile of production to reduce sparse-tail noise). We compare degree-1 and degree-2 fits to assess nonlinearity, and test bandwidth robustness at $0.5\times$, $1\times$, and $2\times$ the rule-of-thumb. In the lagged specification, the lagged production term is first OLS-partialled from wages via the Frisch–Waugh–Lovell (FWL) theorem before the local polynomial is applied, isolating the *current*-year effect.

Conditional quantile regression (Koenker–Bassett) is estimated at $\tau \in \{0.10, 0.50, 0.90\}$, with and without the geology instrument (Chernozhukov–Hansen IVQR). The IV specification requires a rank stability condition: that a county's position in the wage-growth ranking is roughly the same regardless of whether it was fracked, in addition to the exclusion restriction. We flag where the instrument is likely too weak for reliable IVQR inference.

Unconditional RIF regression (Firpo, Fortin and Lemieux, 2009) replaces the dependent variable with its re-centred influence function evaluated at each quantile, then applies OLS. This delivers the effect of a marginal increase in fracking on the actual p10, p50, and p90 of the cross-county wage-growth distribution, providing a direct inequality statement. There is no instrument in the RIF specification; it should be read as associational, not causal.

4. Nonparametric Analysis

4.1 Distributional comparison

Figure 2 shows the kernel density and empirical CDF of demeaned wage growth for high-fracking and no-fracking counties. The key finding is not a rightward shift but a *widening*: the high-fracking density is flatter and heavier-tailed in both directions. The CDFs cross, ruling out first-order stochastic dominance of either distribution over the other. A Kolmogorov–Smirnov test confirms the two distributions are statistically distinguishable. Fracking creates more winners and more losers; it does not simply lift all counties uniformly.

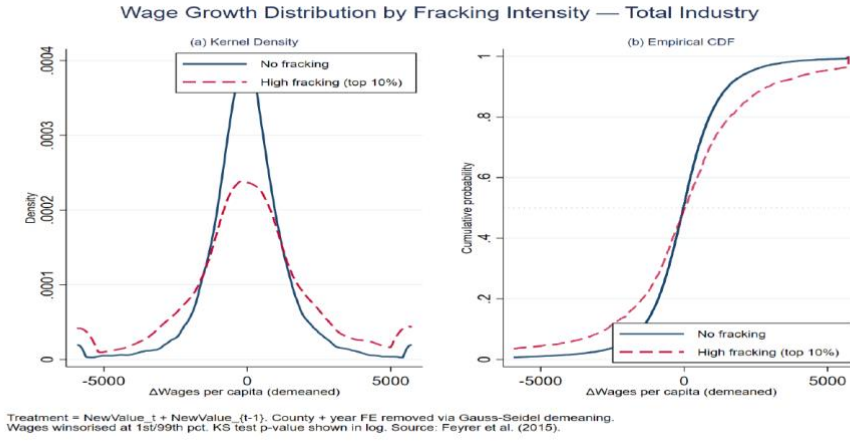


Figure 2. Wage-growth distribution by fracking intensity — Total industry. Kernel density (left) and empirical CDF (right). High-fracking counties (dashed) have fatter tails in both directions; the CDFs cross, ruling out first-order stochastic dominance.

4.2 Linearity check and bandwidth robustness

Figure 3 re-estimates the local-linear fit across three bandwidths to assess sensitivity to the smoothing parameter. The bias-variance trade-off is central to interpreting these results: a bandwidth that is too narrow reduces bias by tracking local variation closely but inflates variance and risks over-fitting to noise; a bandwidth that is too wide suppresses variance at the cost of smoothing over genuine curvature. The $0.5\times$ bandwidth illustrates the undersmoothing extreme: the estimator overfits in the sparse right tail, where few observations anchor the fit, and the resulting curve reflects noise rather than signal. The $1\times$ and $2\times$ bandwidths, by contrast, occupy a more favourable region of this trade-off: both recover the same hump-shaped pattern, wages rising with drilling intensity at low-to-moderate levels, peaking, then declining: suggesting this curvature is not an artefact of bandwidth choice but a stable feature of the conditional expectation function. The OLS linear fit illustrates the opposite failure mode: by imposing a global parametric restriction, it oversmooths entirely, trimming the initial rise and eliminating the right-tail decline and incurring specification bias throughout the support. That the $1\times$ and $2\times$ curves agree despite the change in smoothing provides informal evidence that the chosen bandwidth is neither severely under nor oversmoothed. The recovered nonlinearity is consistent with diminishing, and eventually negative, marginal returns at very high drilling density, plausibly reflecting labour-market congestion and housing cost pressures in heavily drilled areas.

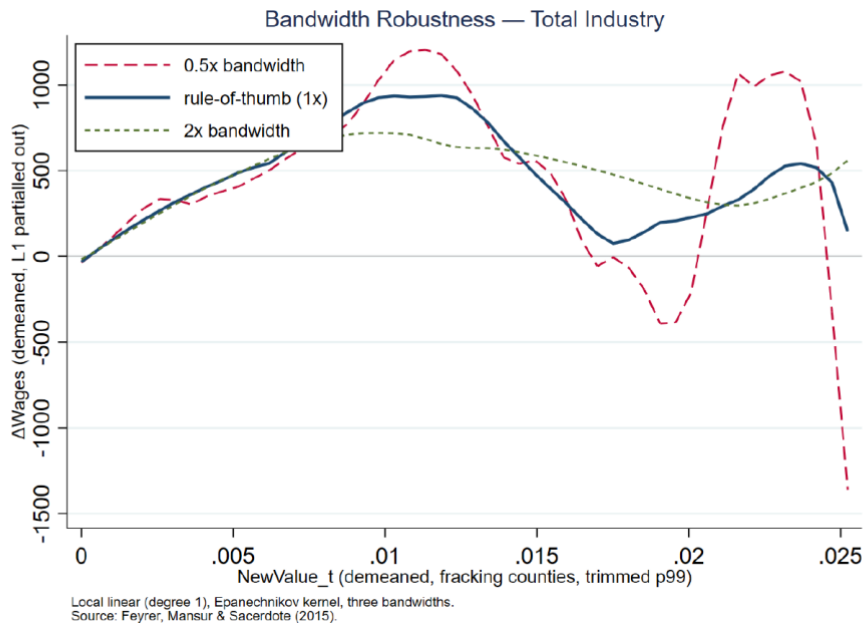


Figure 3. Bandwidth robustness check — Total industry. Local-linear fits at $0.5\times$, $1\times$, and $2\times$ the rule-of-thumb bandwidth. The hump shape is robust at $1\times$ and $2\times$; the $0.5\times$ curve over-fits in the sparse right tail. The OLS linear fit is shown as a dotted reference.

5. Quantile Regression Results

5.1 Conditional quantile process (no instrument)

Figures 4a and 4b plot the conditional quantile regression coefficient on current production across all deciles, alongside the OLS benchmark. For the total economy (Figure 4a), the curve hugs the OLS line ($\sim \$34,000$) throughout, drifting up modestly to $\sim \$39,000$ at Q90, but the confidence band contains the OLS line everywhere. Once we condition on county and year fixed effects, the distributional evidence for the total economy is weak: the conditional effect is roughly homogeneous, consistent with FMS's implicit single-coefficient assumption.

Mining tells a sharper story (Figure 4b). The curve rises from $\sim \$8,000$ at the lowest deciles to a peak near $\$20,000$ around the 70th percentile, then falls back sharply. At the bottom deciles, the coefficient lies *significantly below* the OLS line: the mean overstates the fracking effect for low-growth mining counties. This cross-quantile heterogeneity motivates the IVQR analysis below.

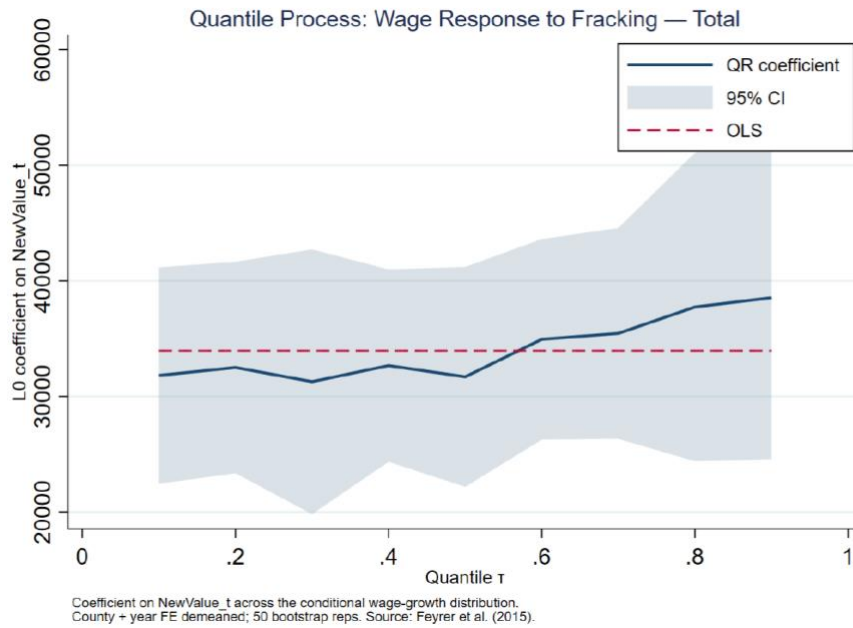


Figure 4a. Conditional quantile process — Total industry. The QR coefficient (solid blue) hugs the OLS line (dashed red) across the distribution; the 95% confidence band contains OLS throughout.

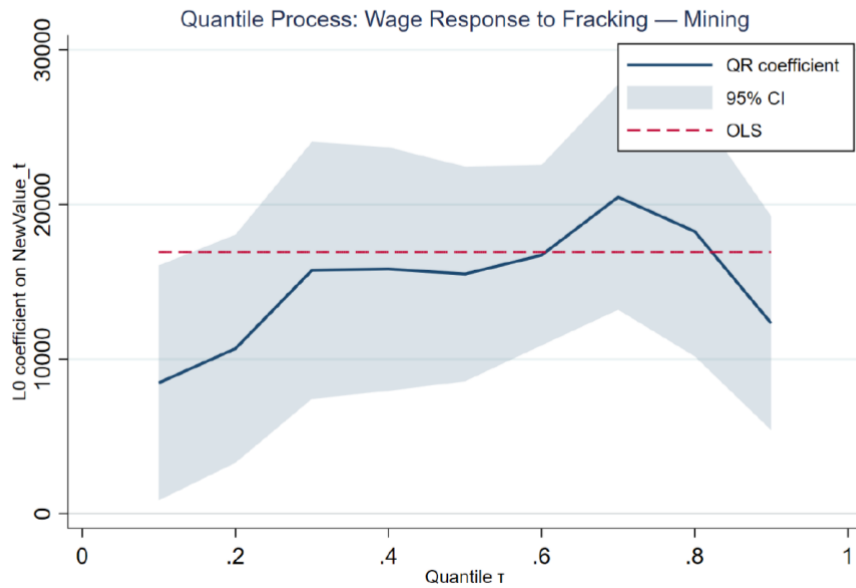


Figure 4b. Conditional quantile process — Mining. The effect peaks near the 70th percentile and falls significantly below OLS at the lowest deciles, indicating the mean overstates the benefit for low-growth counties.

5.2 IV quantile regression

Figure 5 plots the Chernozhukov–Hansen IVQR net effect ($L0 + L1$) by industry and quantile. Two findings stand out. First, instrumentation matters enormously: the IVQR contemporaneous ($L0$) term for Total ($\sim \$86,000$ – $\$94,000$ across quantiles) is roughly double the no-IV estimate, confirming that OLS substantially understates the causal effect. Second, the *net* effect (current plus lagged) *declines* across quantiles, from $\sim \$84,000$ at Q10 to $\sim \$58,000$ at Q90, because the lagged give-back grows sharply at the top of the distribution (near zero at Q10 but $\sim \$37,000$ at Q90). High-growth counties enjoy a larger immediate boom but absorb a sharper correction in the following year: a boom–bust signature the mean cannot detect. For Mining and Transport the effects are large and cleanly significant; for Education, Government, and Construction the instrument is weak and the wide confidence intervals should not be over-interpreted.

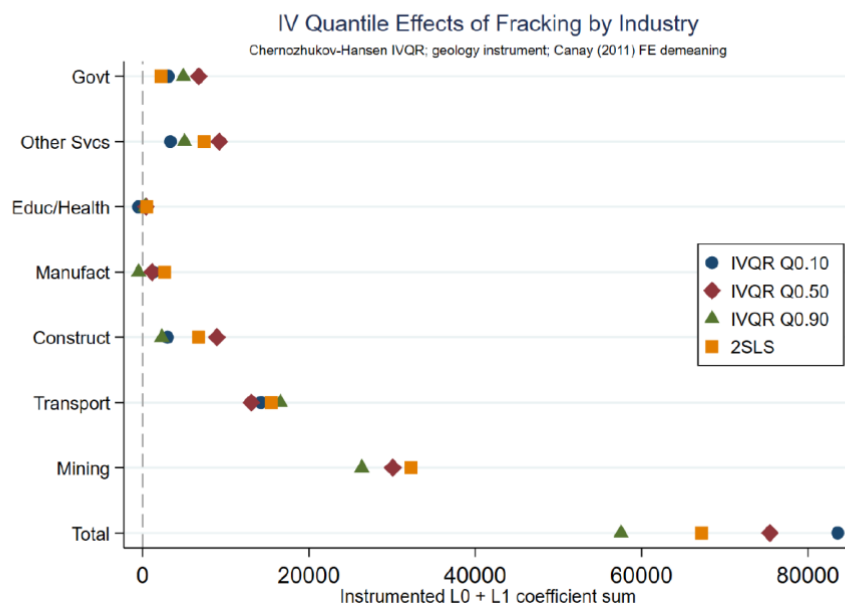


Figure 5. IV quantile effects ($L0 + L1$ net) by industry and quantile — Chernozhukov–Hansen IVQR with geology instrument. Total markers fan out: the net effect is largest at Q10 (circle) and smallest at Q90 (triangle). Mining and Transport are the cleanest sectors; wide intervals for Education, Government, and Construction reflect weak-instrument territory.

5.3 Unconditional RIF regression

Figure 6 and Table 1 present the headline result. For Total, the RIF effect is \$8,246 at Q10, \$4,765 at Q50, and \$31,304 at Q90 — all significant at the 1 per cent level. The 90th-percentile effect is roughly six times the median. Fracking disproportionately lifts the upper tail of the cross-county wage distribution, widening spatial inequality. Mining, Transport, and Other Services share the same top-heavy profile. Manufacturing is mildly negative, consistent with Dutch Disease crowding-out.

The contrast with earlier estimates is clarifying. The conditional quantile regression without an instrument reproduces the OLS mean across the distribution ($\sim \$31,000$ – $\$39,000$). Adding the instrument raises the level sharply. Switching to the unconditional distribution *relocates* the effect to the upper tail: the median and lower-decile impacts are small; the action is at Q90. These three estimands answer genuinely different questions and yield genuinely different answers; none is simply a scaled version of another (see Table 1).

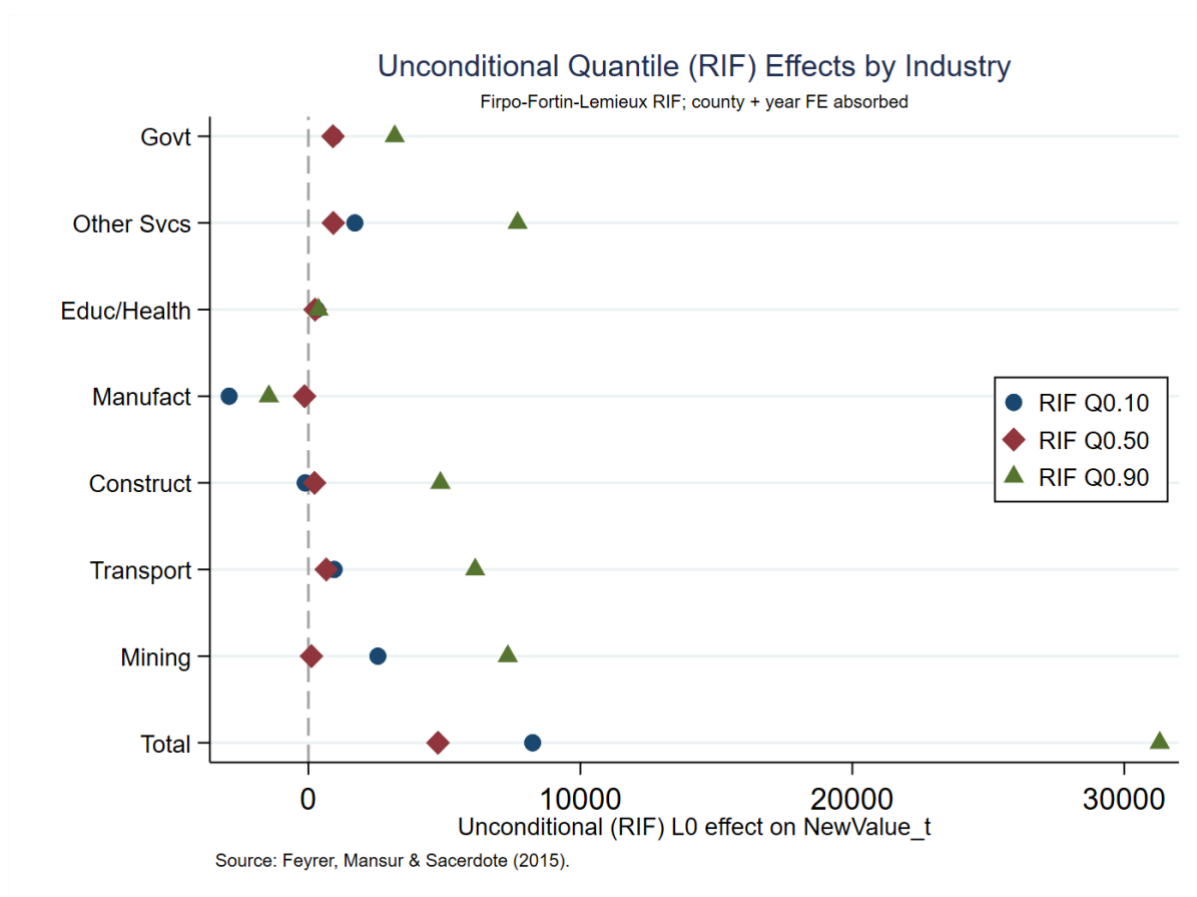


Figure 6. Unconditional RIF effects by industry and quantile (Firpo–Fortin–Lemieux). The Q90 markers (triangles) almost always sit far to the right of Q10 (circles) and Q50 (diamonds), particularly for Total, Mining, and Transport — fracking raises the upper tail of the wage distribution markedly more than the middle or bottom.

Table 1: Estimated wage effects by specification, Total industry (L0 term, \$ per worker per \$1m production)

Specification	Q0.10	Q0.50	Q0.90
OLS (FMS benchmark)	33,957	33,957	33,957
Conditional QR (no IV)	31,812	31,694	38,547
IVQR — Chernozhukov–Hansen	86,457***	92,055***	94,448***
Unconditional RIF (no IV)	8,246***	4,765***	31,304***

Notes: All figures in US dollars per worker per million dollars of new production. *** $p < 0.01$.

6. Discussion

6.1 What the extensions add

Our nonparametric evidence challenges FMS's implicit linearity assumption. The local polynomial results consistently reveal a hump-shaped treatment response: returns to drilling intensity rise at first and then diminish and eventually reverse. The paper's single coefficient averages across these different regimes. This matters for extrapolation: the FMS estimate is most reliable near the mean of the production distribution and should not be applied to very high-intensity counties without qualification.

The IVQR results confirm the paper's central message: instrumentation roughly doubles the estimated effect, whilst revealing a boom–bust asymmetry. The net effect (current plus lagged) is largest for counties in the bottom decile of wage growth and smallest for those at the top. This is not because fracking fails to help high-growth counties; the contemporaneous effect is larger at Q90. The mechanism is the sharper give-back: high-growth counties experience a larger initial shock but also a stronger mean reversion in the following year, consistent with the rapid dissipation of mining wages that FMS document in their impulse-response analysis. A policy aimed at generating *durable* wage gains would therefore target mid-distribution rather than top of distribution counties.

The RIF result is the sharpest departure from the original paper. FMS document large average gains but cannot say who captures them. Our RIF estimates show that the gains are concentrated at the upper tail of the cross-county distribution: fracking amplifies spatial wage inequality rather than compressing it. Any welfare evaluation that applies greater weight to low-wage counties will therefore assess the boom less favourably than FMS's mean estimate implies.

6.2 Caveats

Our IVQR point estimates for Total are broadly consistent with FMS's 2SLS benchmark (\$86K–\$94k vs \$79,751), though some industry level figures diverge. The likely cause is our just-identified demeaned specification, which lacks the richer set of controls and two-way-clustered standard errors of FMS's `xtivreg2` implementation. We accordingly emphasise the across-quantile pattern rather than the absolute dollar levels.

The RIF estimator carries no instrument. In principle, counties with pre-existing industrial capacity or transport infrastructure may simultaneously attract fracking activity and occupy the upper tail of the wage-growth distribution, confounding the RIF estimates. The consistency between the RIF story and the causal IVQR story, with both pointing to concentration at the top, is reassuring but not conclusive.

Our do files also produced a "spillover decay" figure that rises from roughly \$35,000 at the county level to approximately \$250,000 at 100 miles before plateauing. That figure uses cumulative disks, everything within X miles rather than annular rings, so it traces the accumulation of effects as the radius expands, which is precisely the paper's own spatial-reach finding. Presenting it as a new extension would misrepresent it, and we therefore exclude it from the main body of the paper.

7. Conclusion

Feyrer, Mansur and Sacerdote (2017) establish that fracking generates large average causal wage effects propagating up to 100 miles from drilling sites, with roughly two-thirds of the initial increase persisting after two years. Our extensions show that the average conceals three important features. The dose–response is hump-shaped rather than linear, with the marginal wage return diminishing at very high drilling intensity. The causal effect varies across the wage-growth distribution: high-growth counties gain most on impact but correct most sharply in the following year, a boom–bust pattern the mean obscures. Most distinctively, fracking *widens* the cross-county wage distribution: the unconditional effect at the 90th percentile is over six times the median, implying that the boom amplifies spatial inequality rather than compressing it.

These findings do not overturn FMS; they enrich it. A complete welfare assessment of a localised resource boom requires distributional evidence alongside the conditional mean, which have been provided by our extensions.

References

- Canay, I.A. (2011). "A simple approach to quantile regression for panel data." *The Econometrics Journal*, 14(3), 368–386.
- Chernozhukov, V. and Hansen, C. (2005). "An IV model of quantile treatment effects." *Econometrica*, 73(1), 245–261.
- Feyrer, J., Mansur, E.T. and Sacerdote, B. (2017). "Geographic dispersion of economic shocks: evidence from the fracking revolution." *American Economic Review*, 107(4), 1313–1334.
- Firpo, S., Fortin, N.M. and Lemieux, T. (2009). "Unconditional quantile regressions." *Econometrica*, 77(3), 953–973.
- Koenker, R. and Bassett, G. (1978). "Regression quantiles." *Econometrica*, 46(1), 33–50.

Appendix A: Full Per-Industry Coefficient Tables

Table A1: RIF unconditional quantile estimates by industry

Industry	Term	Q0.10	Q0.50	Q0.90
Total	L0 (current)	8,246***	4,765***	31,304***
	L1 (lag)	−5,670**	−2,548**	−9,324*
Mining	L0 (current)	2,555***	111***	7,333***
	L1 (lag)	−1,964**	−60	−4,281**
Transport	L0 (current)	948**	658***	6,133***
	L1 (lag)	−43	−207	−1,421
Construction	L0 (current)	−118	226**	4,855***
	L1 (lag)	913	12	−4,457***
Manufacturing	L0 (current)	−2,915*	−141	−1,454*
	L1 (lag)	3,639*	254	22
Educ/Health	L0 (current)	303	245*	371
	L1 (lag)	−86	−77	−293
Other Services	L0 (current)	1,710**	917***	7,692***
	L1 (lag)	−1,092	−56	1,112
Government	L0 (current)	930	904***	3,174*
	L1 (lag)	1,648	−38	2,443

*Firpo–Fortin–Lemieux RIF estimator; Canay (2011) FE demeaning; bootstrap standard errors (100 replications). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

Table A2: IVQR estimates by industry

Industry	Term	Q0.10	Q0.50	Q0.90	2SLS
Total	L0	86,457***	92,055***	94,448***	79,751***
	L1	−2,884	−16,630	−36,918***	−12,534
Transport	L0	17,203	19,099***	20,495***	19,691***
	L1	−2,961	−6,028***	−3,929	−4,211
Construction	L0	3,562**	9,516	2,375***	10,070***
	L1	−569	−585	−59	−3,338
Manufacturing	L0	−3,846**	−1,873***	−1,451	−2,455

Industry	Term	Q0.10	Q0.50	Q0.90	2SLS
Educ/Health	L1	5,145***	3,015***	965	5,065*
	L0	640	1,091***	813	1,682
Other Services	L1	-1,150	-686*	-490	-1,203
	L0	4,265***	10,280	10,367***	6,998***
Government	L1	-929	-1,067	-5,327	408
	L0	-1,333	3,111**	3,338***	15,122
	L1	4,440***	3,638***	1,556	-12,894

Chernozhukov–Hansen IVQR; geology instrument; Canay (2011) FE demeaning. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.